

# Python Programming: Lab Activity 4

**Due Date:** 7/03/2025

## Objective

This lab focuses on **Functions, loops, Lists and Modules** in Python. The exercises are designed to enhance your understanding of:

- Defining and using functions to perform specific tasks
- Using loops for repeated actions
- Handling user input and performing basic operations on it
- Working with the random and math module to generate random values
- Creating and manipulating 2D lists (matrices) to represent structured data

## Assignment Tasks

### Q1: Find the largest number

- Define a function called `max_of_three()` that prompts users to enter three integer inputs, and return the largest number on the screen. (4 points)

### Q2: Check palindrome

- Define a function `isPalindrome()` that takes an integer `num1` from the user and checks whether the number is a palindrome (a number that reads the same forwards and backwards). The function should return `True` if the number is a palindrome, and `False` otherwise. For example, if the input is 121, the output should be `True`, and if the input is 123, the output should be `False` (4 points)

### Q3: Random number guessing game

- Write a program that generates a random number in the range of 1 through 100, and asks the user to guess what the number is. If the user's guess is higher than the random number, the program should display "Too high, try again". If the user's guess is lower than the random number, the program should display "Too low, try again". If the user guesses the number, the application should congratulate the user and generate a new random number so the game can start over. (6 points)

### Q4: Find the square root

- Write a Python function that takes a number from the user and prints its square root using the `math` module. If the user enters a negative number, display an error message (5 points)

### Q5: Create a Matrix

- Write a Python program that creates a 2D list (a list of lists) to represent a matrix. Begin by asking the user to input the number of rows and columns for the matrix. Ensure that the inputs are positive integers. Then, create an empty matrix with the specified number of rows and columns. Use nested loops to fill the matrix by prompting the user to enter values for each cell in the matrix. After the matrix is filled, print the matrix to display the values in a readable format, where each row appears on a new line (6 points)

## Submission Guidelines

1. Submit the provided file lab4.py with existing template code.
2. **Do not make any changes in the template code provided.** Please only fill the said places in the file
3. Ensure that your program runs correctly before submission.
4. Please do not rename the file.

## Grading Criteria

- **Correctness:** Output must match the expected results exactly.
- **Completeness:** All tasks must be completed as per instructions.
- **Efficiency:** Code should be optimized where possible.
- **Readability:** Use proper indentation and comments to explain logic.
- **Originality:** Avoid plagiarism—write your own code.